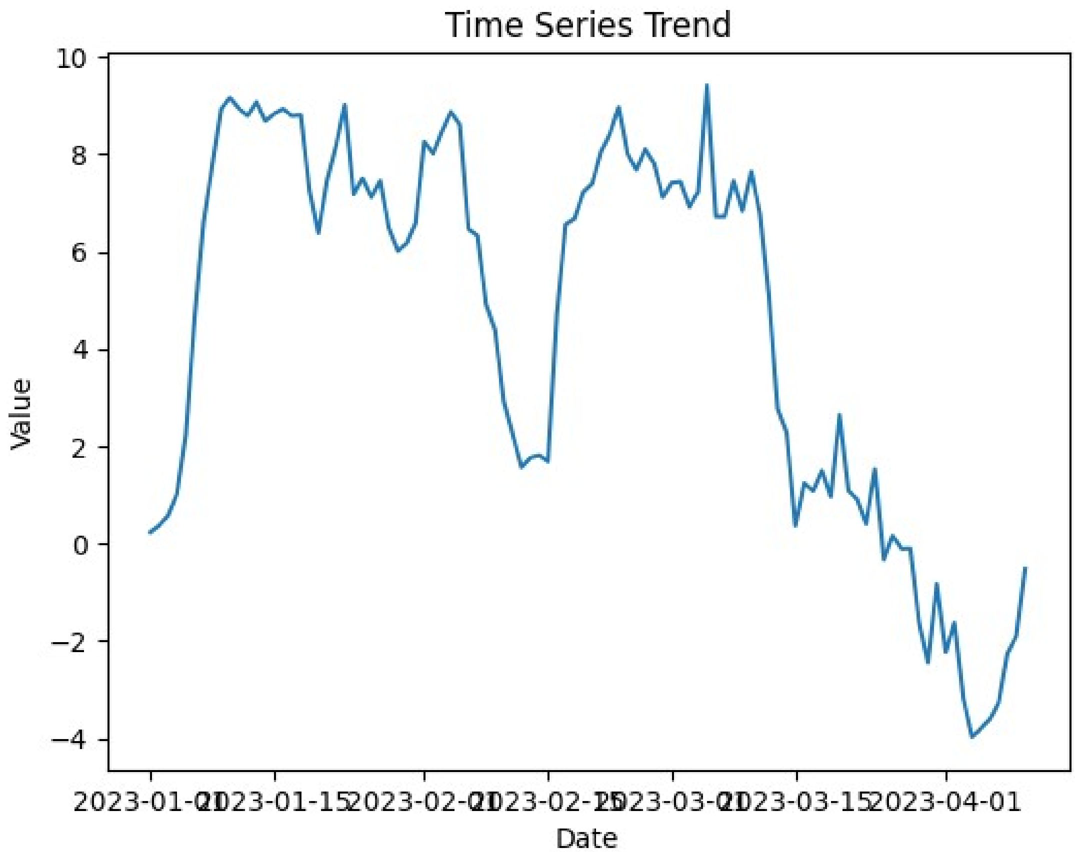
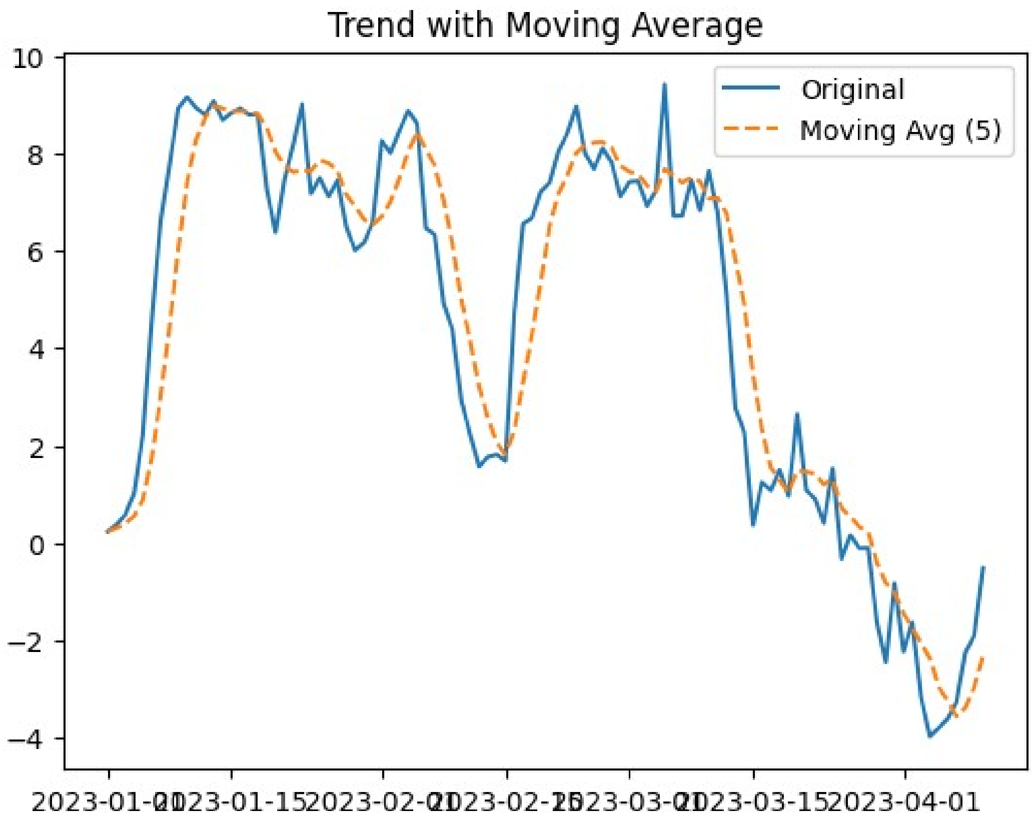
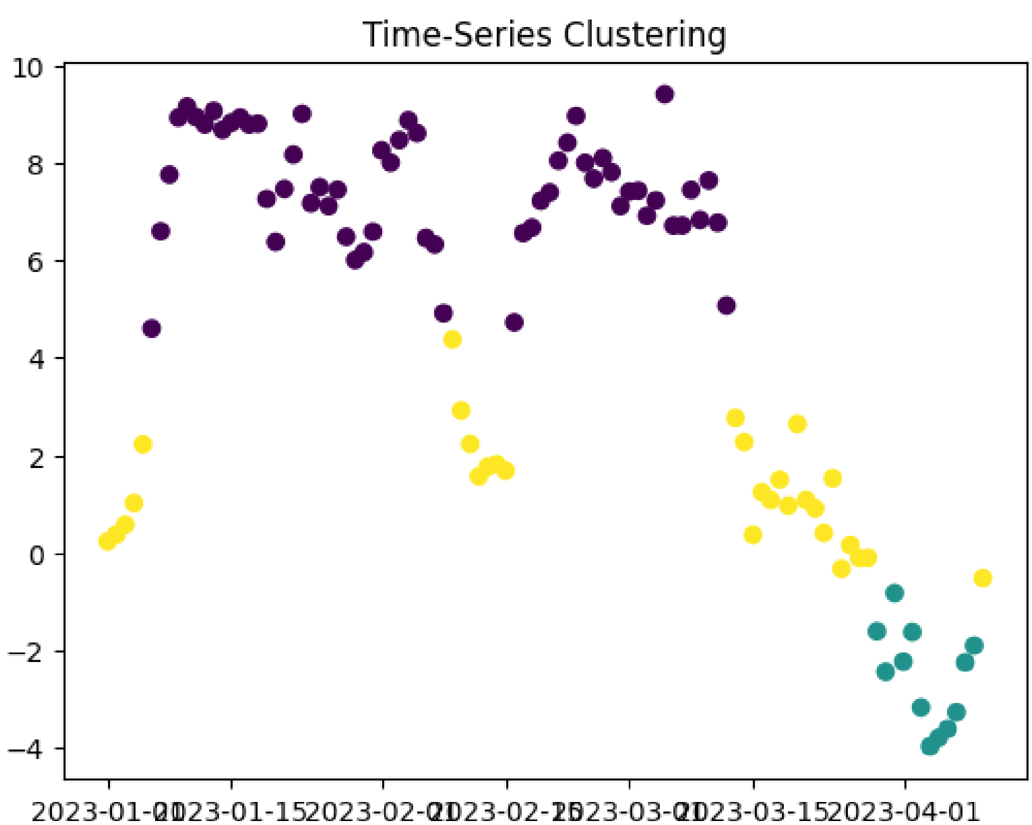
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Step 1: Create sample time-series data
dates = pd.date_range(start='2023-01-01', periods=100)
data = np.cumsum(np.random.randn(100))  # cumulative sum (trend-like)

df = pd.DataFrame({'Date': dates, 'Value': data})
df.set_index('Date', inplace=True)

# Step 2: Plot time-series (Trend Analysis)
plt.figure()
plt.plot(df.index, df['Value'])
plt.title("Time Series Trend")
plt.xlabel("Date")
plt.ylabel("Value")
plt.show()

# Step 3: Moving Average (Temporal Pattern)
df['Moving_Avg'] = df['Value'].rolling(window=5, min_periods=1).mean()

plt.figure()
plt.plot(df.index, df['Value'], label='Original')
plt.plot(df.index, df['Moving_Avg'], label='Moving Avg (5)',
linestyle='--')
plt.legend()
plt.title("Trend with Moving Average")
plt.show()

# Step 4: Clustering time-based values
# Using only values for clustering
kmeans = KMeans(n_clusters=3, random_state=0)
df['Cluster'] = kmeans.fit_predict(df[['Value']])

plt.figure()
plt.scatter(df.index, df['Value'], c=df['Cluster'])
plt.title("Time-Series Clustering")
plt.show()

print(df.head())
```



Time Series Trend
Value
10
8
6
4
2
0
−2
−4
2023-01-01
2023-01-15
2023-02-01
2023-02-15
2023-03-01
2023-03-15
2023-04-01
Date


Trend with Moving Average
Original
Moving Avg (5)
10
8
6
4
2
0
−2
−4
2023-01-01
2023-01-15
2023-02-01
2023-02-15
2023-03-01
2023-03-15
2023-04-01


Time-Series Clustering
10
8
6
4
2
0
−2
−4
2023-01-01
2023-01-15
2023-02-01
2023-02-15
2023-03-01
2023-03-15
2023-04-01

Date	Value	Moving_Avg	Cluster
2023-01-01	0.238657	0.238657	2
2023-01-02	0.375473	0.307065	2
2023-01-03	0.580566	0.398232	2
2023-01-04	1.022604	0.554325	2
2023-01-05	2.230621	0.889584	2